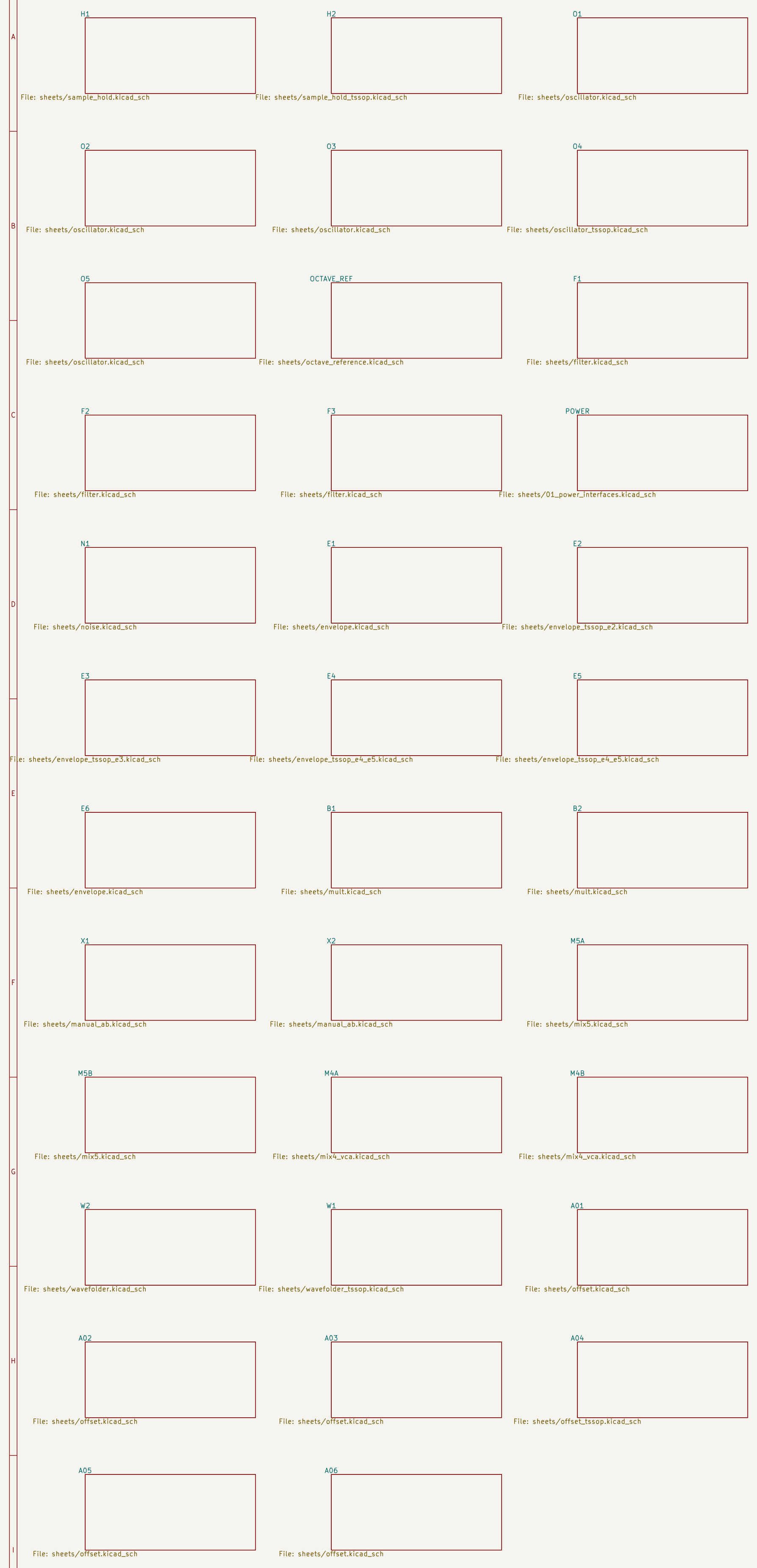
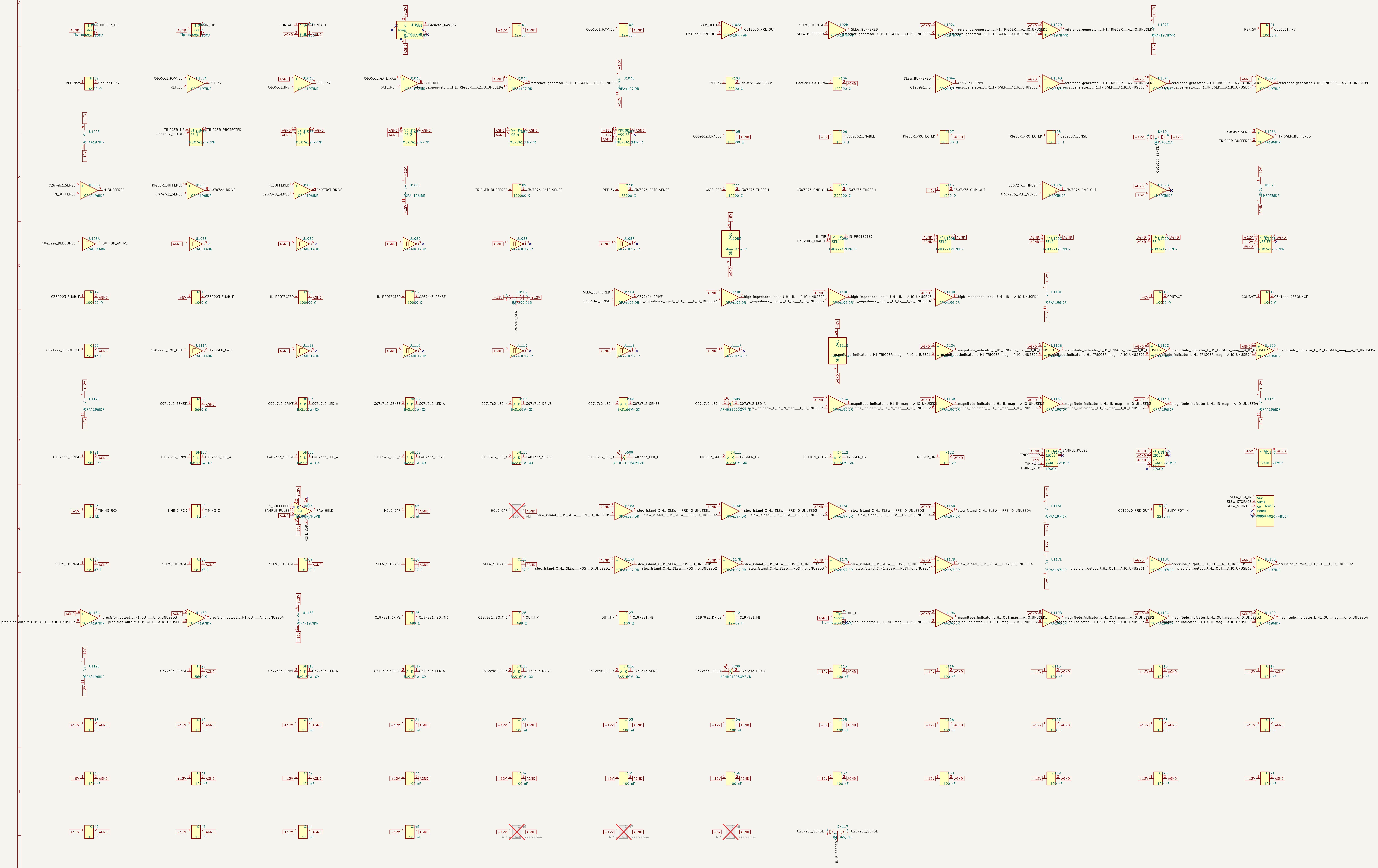

H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold_tssop.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator_tssop.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope_tssop_e2.kicad_sch
E3
File: sheets/envelope_tssop_e3.kicad_sch
E4
File: sheets/envelope_tssop_e4_e5.kicad_sch
E5
File: sheets/envelope_tssop_e4_e5.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder_tssop.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset_tssop.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

Sheet: /
File: zudo–osc–hole–field.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 1/36



Sheet: /H1/
File: sample_hold.kicad_sch

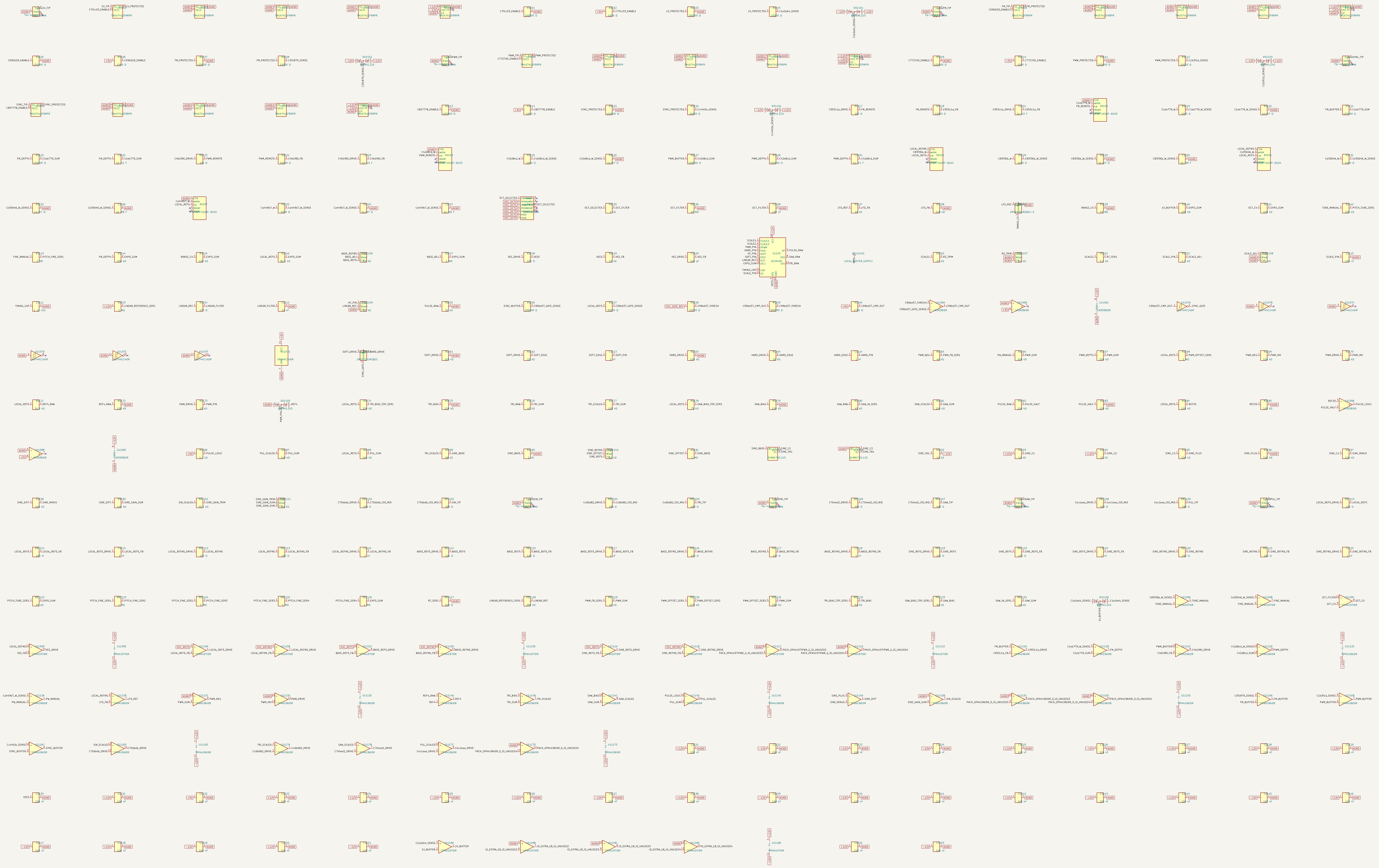
Title:

Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 2/36

Sheet: /H2/
File: sample_hold_tssop.kicad_sch

Title:

Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 3/36



Sheet: /01/
File: oscillator.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 4/36

Sheet: /02/
File: oscillator.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/36

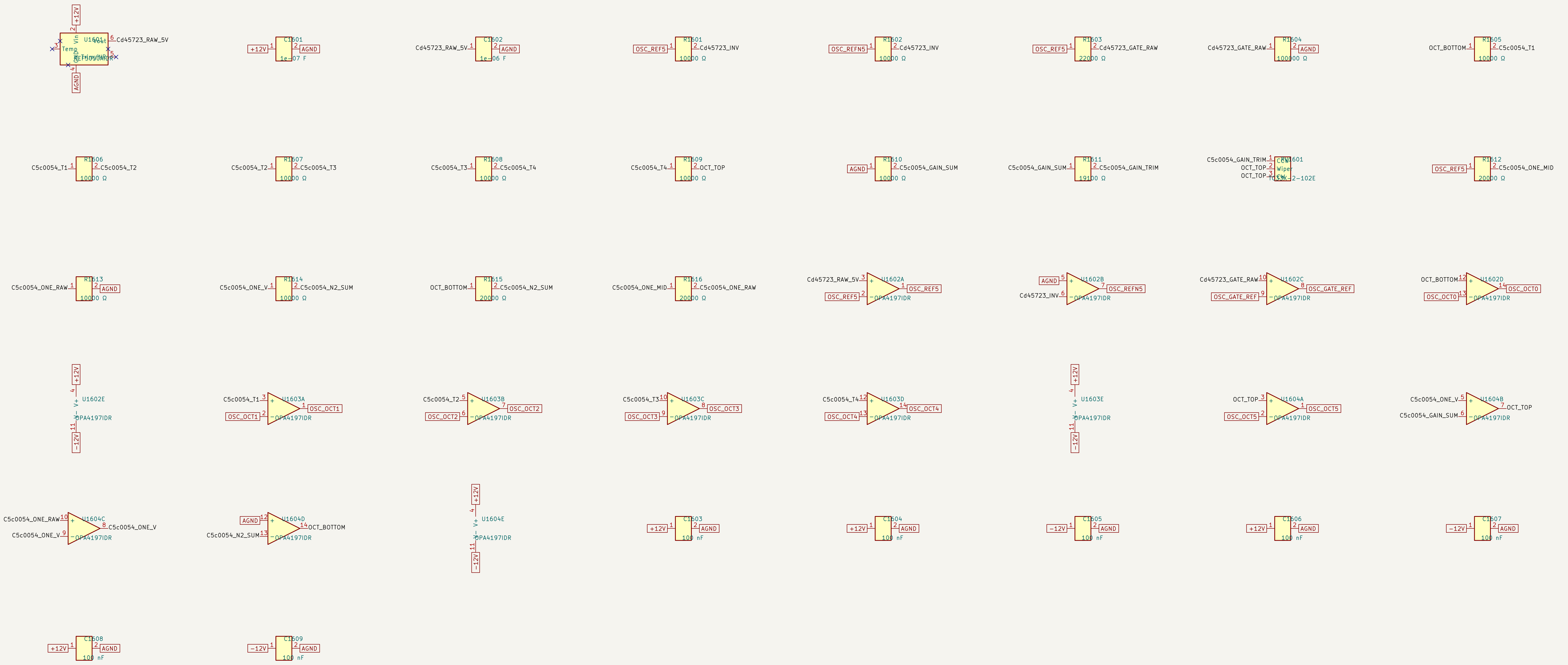
Sheet: /03/
File: oscillator.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 6/36

Sheet: /04/
File: oscillator_tssop.kicad_sch
Title:
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 7/36

Sheet: /05/
File: oscillator.kicad_sch
Title:
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 8/36



Sheet: /OCTAVE_REF/
File: octave_reference.kicad_sch

Title:

Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 9/36

Sheet: /F1/
File: filter.kicad_sch

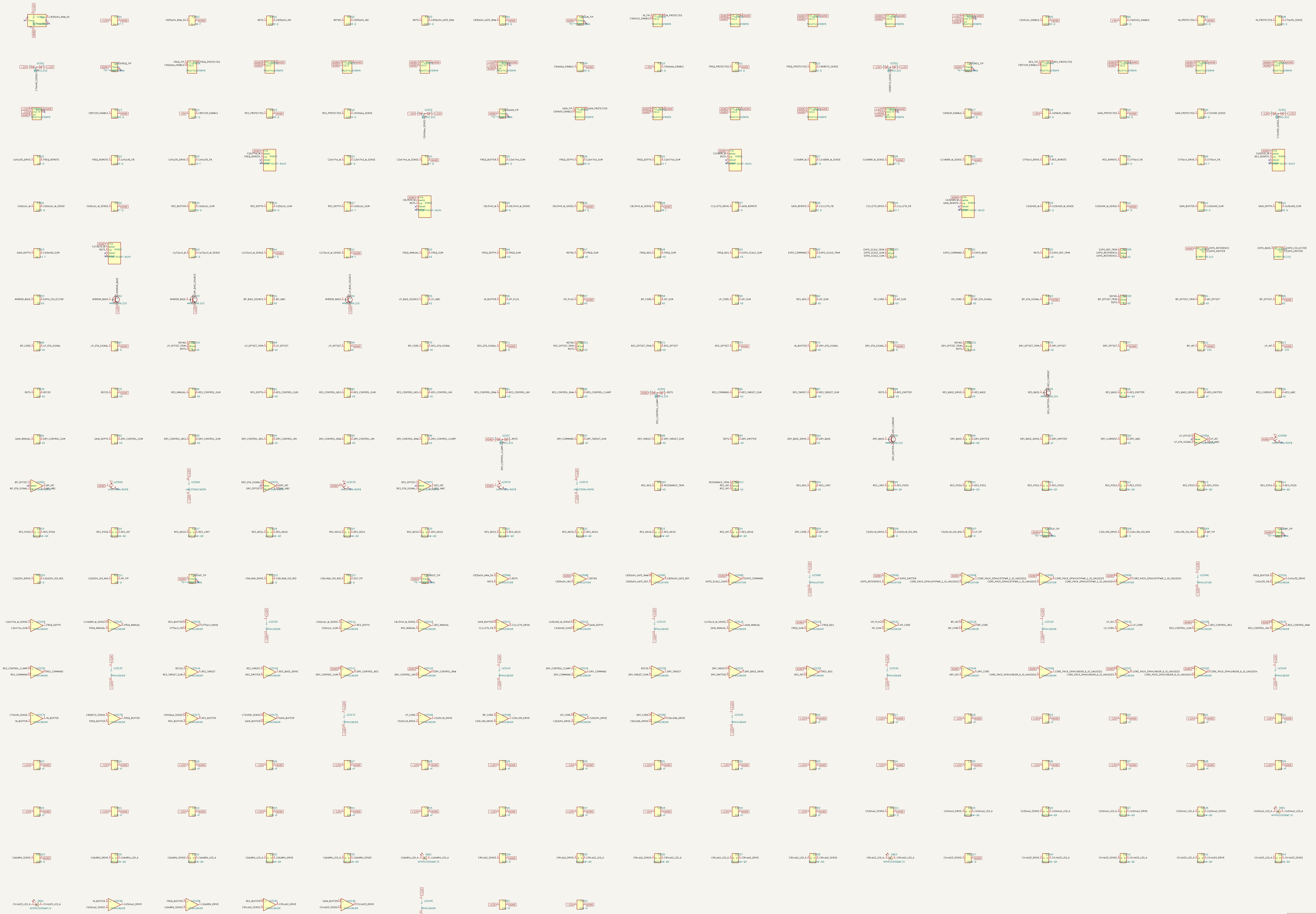
Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 10/36

Sheet: /F2/
File: filter.kicad_sch

Title:

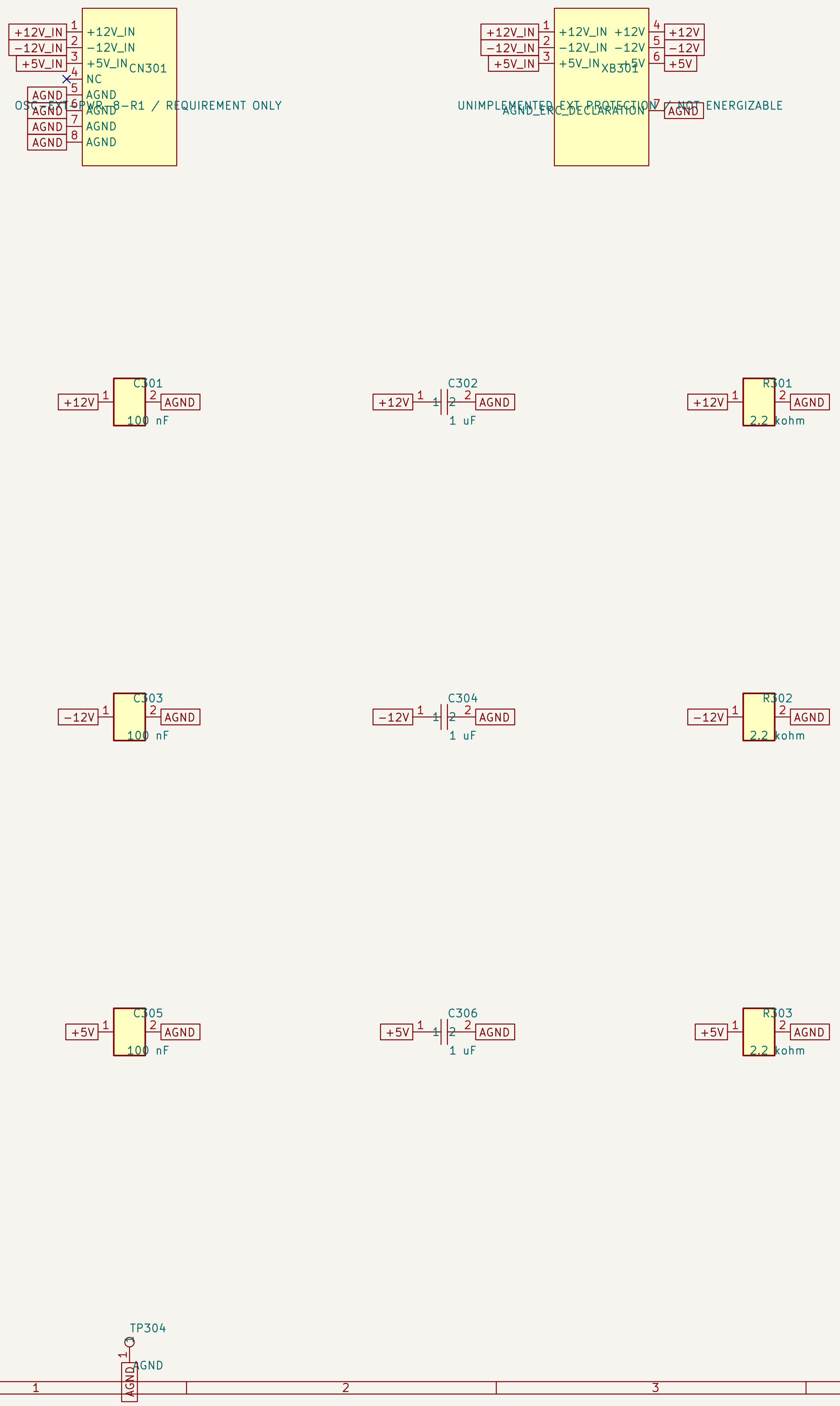
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 11/36



Sheet: /F3/
File: filter.kicad_sch

Title:

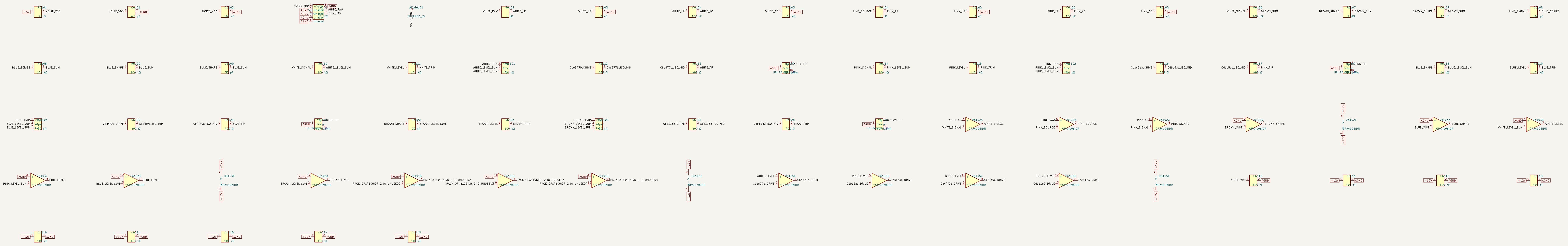
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 12/36



Sheet: /POWER/
File: 01_power_interfaces.kicad_sch

Title:

Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 13/36


+5V R6101 NOISE_VDD 22 Ω
NOISE_VDD C6101 AGND 4.7 µF
NOISE_VDD C6102 AGND 100 nF
NOISE_VDD U6101 +5V 0V AGND WHITE_OUT PINK_OUT Unused Unused NOISE2 WHITE_RAW PINK_RAW
#FLG6101 NOISE_VDD FILTERED_5V
WHITE_RAW R6102 WHITE_LP 1 kΩ
WHITE_LP C6103 AGND 10 nF
WHITE_LP C6104 WHITE_AC 100 nF
WHITE_AC R6103 AGND 100 kΩ
PINK_SOURCE R6104 PINK_LP 1 kΩ
PINK_LP C6105 AGND 10 nF
PINK_LP C6106 PINK_AC 100 nF
PINK_AC R6105 AGND 100 kΩ
WHITE_SIGNAL R6106 BROWN_SUM 100 kΩ
BROWN_SHAPE R6107 BROWN_SUM 1 MΩ
BROWN_SHAPE C6107 BROWN_SUM 10 nF
PINK_SIGNAL C6108 BLUE_SERIES 100 pF
BLUE_SERIES R6108 BLUE_SUM 100 kΩ
BLUE_SHAPE R6109 BLUE_SUM 100 kΩ
BLUE_SHAPE C6109 BLUE_SUM 33 pF
WHITE_SIGNAL R6110 WHITE_LEVEL_SUM 100 kΩ
WHITE_LEVEL R6111 WHITE_TRIM 100 kΩ
WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM RV6101 CCW Wiper CW 10 kΩ
Cbe877b_DRIVE R6112 Cbe877b_ISO_MID 499 Ω
Cbe877b_ISO_MID R6113 WHITE_TIP 499 Ω
AGND S Tip Sleeve TN Tip−normal WHITE_TIP
PINK_SIGNAL R6114 PINK_LEVEL_SUM 100 kΩ
PINK_LEVEL R6115 PINK_TRIM 100 kΩ
PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM RV6102 CCW Wiper CW 10 kΩ
Cdbc5aa_DRIVE R6116 Cdbc5aa_ISO_MID 499 Ω
Cdbc5aa_ISO_MID R6117 PINK_TIP 499 Ω
AGND S Tip Sleeve TN Tip−normal PINK_TIP
BLUE_SHAPE R6118 BLUE_LEVEL_SUM 10 kΩ
BLUE_LEVEL R6119 BLUE_TRIM 100 kΩ
BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM RV6103 CCW Wiper CW 10 kΩ
Ce44f9a_DRIVE R6120 Ce44f9a_ISO_MID 499 Ω
Ce44f9a_ISO_MID R6121 BLUE_TIP 499 Ω
AGND S Tip Sleeve TN Tip−normal BLUE_TIP
BROWN_SHAPE R6122 BROWN_LEVEL_SUM 20 kΩ
BROWN_LEVEL R6123 BROWN_TRIM 100 kΩ
BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM RV6104 CCW Wiper CW 10 kΩ
Cde1183_DRIVE R6124 Cde1183_ISO_MID 499 Ω
Cde1183_ISO_MID R6125 BROWN_TIP 499 Ω
AGND S Tip Sleeve TN Tip−normal BROWN_TIP
WHITE_AC WHITE_SIGNAL U6102A OPA4196IDR WHITE_SIGNAL
PINK_RAW PINK_SOURCE U6102B OPA4196IDR PINK_SOURCE
PINK_AC PINK_SIGNAL U6102C OPA4196IDR PINK_SIGNAL
AGND BROWN_SUM U6102D OPA4196IDR BROWN_SHAPE
+12V −12V U6102E OPA4196IDR
AGND BLUE_SUM U6103A OPA4196IDR BLUE_SHAPE
AGND WHITE_LEVEL_SUM U6103B OPA4196IDR WHITE_LEVEL
AGND PINK_LEVEL_SUM U6103C OPA4196IDR PINK_LEVEL
AGND BLUE_LEVEL_SUM U6103D OPA4196IDR BLUE_LEVEL
+12V −12V U6103E OPA4196IDR
AGND BROWN_LEVEL_SUM U6104A OPA4196IDR BROWN_LEVEL
AGND PACK_OPA4196IDR_2_IO_UNUSED2 U6104B OPA4196IDR PACK_OPA4196IDR_2_IO_UNUSED2
AGND PACK_OPA4196IDR_2_IO_UNUSED3 U6104C OPA4196IDR PACK_OPA4196IDR_2_IO_UNUSED3
AGND PACK_OPA4196IDR_2_IO_UNUSED4 U6104D OPA4196IDR PACK_OPA4196IDR_2_IO_UNUSED4
+12V −12V U6104E OPA4196IDR
WHITE_LEVEL Cbe877b_DRIVE U6105A OPA4196IDR Cbe877b_DRIVE
PINK_LEVEL Cdbc5aa_DRIVE U6105B OPA4196IDR Cdbc5aa_DRIVE
BLUE_LEVEL Ce44f9a_DRIVE U6105C OPA4196IDR Ce44f9a_DRIVE
BROWN_LEVEL Cde1183_DRIVE U6105D OPA4196IDR Cde1183_DRIVE
+12V −12V U6105E OPA4196IDR
NOISE_VDD C6110 AGND 100 nF
+12V C6111 AGND 100 nF
−12V C6112 AGND 100 nF
+12V C6113 AGND 100 nF
−12V C6114 AGND 100 nF
+12V C6115 AGND 100 nF
−12V C6116 AGND 100 nF
+12V C6117 AGND 100 nF
−12V C6118 AGND 100 nF

Sheet: /N1/
File: noise.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 14/36

Sheet: /E1/
File: envelope.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 15/36

Sheet: /E2/
File: envelope_tssop_e2.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 16/36

Sheet: /E3/
File: envelope_tssop_e3.kicad_sch
Title:
Size: A0 Date:
KiCad E.D.A. 10.0.6
Rev:
Id: 17/36

Sheet: /E4/
File: envelope_tssop_e4_e5.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 18/36

Sheet: /E5/
File: envelope_tssop_e4_e5.kicad_sch

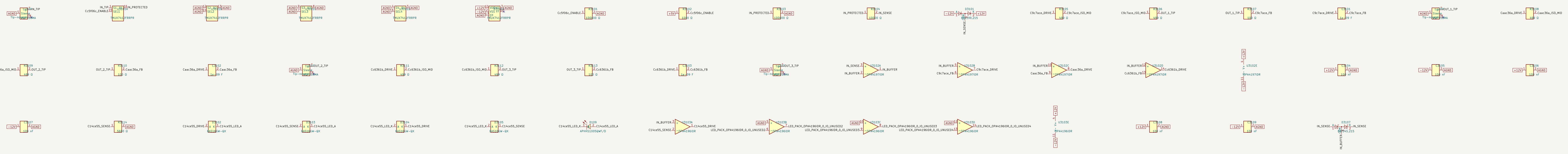
Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 19/36

Sheet: /E6/
File: envelope.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 20/36

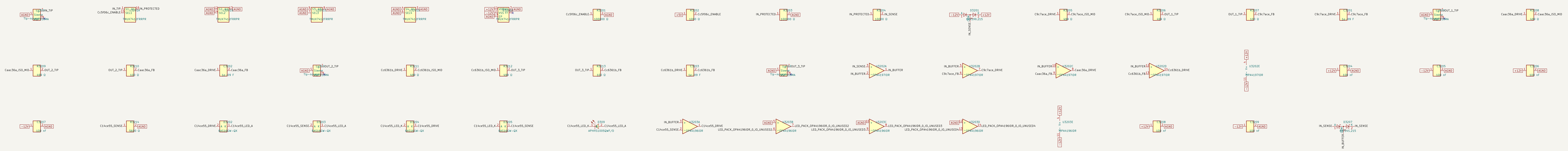

AGND S Tip Sleeve TN IN_TIP Tip-normal WQP518MA
IN_TIP 1 S1 D1 16 IN_PROTECTED
Cc5f06c_ENABLE 15 SEL1 U3101A TMUX7412FRRPR
AGND 12 S2 D2 13 AGND
AGND 14 SEL2 U3101B TMUX7412FRRPR
AGND 9 S3 D3 8 AGND
AGND 7 SEL3 U3101C TMUX7412FRRPR
AGND 4 S4 D4 5 AGND
AGND 6 SEL4 U3101D TMUX7412FRRPR
+12V 11 VDD GND 3 AGND
-12V 2 VSS FF 10
AGND 17 EP TMUX7412FRRPR
Cc5f06c_ENABLE 1 R3101 2 AGND 100000 Ω
+5V 1 R3102 2 Cc5f06c_ENABLE 1000 Ω
IN_PROTECTED 1 R3103 2 AGND 100000 Ω
IN_PROTECTED 1 R3104 2 IN_SENSE 10000 Ω
-12V A 1 D3101 K 2 +12V BAV199,215 COM IN_SENSE
C9c7ace_DRIVE 1 R3105 2 C9c7ace_ISO_MID 499 Ω
C9c7ace_ISO_MID 1 R3106 2 OUT_1_TIP 499 Ω
OUT_1_TIP 1 R3107 2 C9c7ace_FB 100 Ω
C9c7ace_DRIVE 1 C3101 2 C9c7ace_FB 1e-09 F
AGND S Tip Sleeve TN OUT_1_TIP Tip-normal WQP518MA
Caac36a_DRIVE 1 R3108 2 Caac36a_ISO_MID 499 Ω
Caac36a_ISO_MID 1 R3109 2 OUT_2_TIP 499 Ω
OUT_2_TIP 1 R3110 2 Caac36a_FB 100 Ω
Caac36a_DRIVE 1 C3102 2 Caac36a_FB 1e-09 F
AGND S Tip Sleeve TN OUT_2_TIP Tip-normal WQP518MA
Cc6361b_DRIVE 1 R3111 2 Cc6361b_ISO_MID 499 Ω
Cc6361b_ISO_MID 1 R3112 2 OUT_3_TIP 499 Ω
OUT_3_TIP 1 R3113 2 Cc6361b_FB 100 Ω
Cc6361b_DRIVE 1 C3103 2 Cc6361b_FB 1e-09 F
AGND S Tip Sleeve TN OUT_3_TIP Tip-normal WQP518MA
IN_SENSE 3 + U3102A 1 IN_BUFFER
IN_BUFFER 2 − OPA4197IDR
IN_BUFFER 5 + U3102B 7 C9c7ace_DRIVE
C9c7ace_FB 6 − OPA4197IDR
IN_BUFFER 10 + U3102C 8 Caac36a_DRIVE
Caac36a_FB 9 − OPA4197IDR
IN_BUFFER 12 + U3102D 14 Cc6361b_DRIVE
Cc6361b_FB 13 − OPA4197IDR
+12V 4 V+ U3102E V− 11 -12V OPA4197IDR
+12V 1 C3104 2 AGND 100 nF
-12V 1 C3105 2 AGND 100 nF
+12V 1 C3106 2 AGND 100 nF
-12V 1 C3107 2 AGND 100 nF
C14ce55_SENSE 1 R3114 2 AGND 5600 Ω
C14ce55_DRIVE 2 A K 1 C14ce55_LED_A D3102 BAS16GW-QX
C14ce55_SENSE 2 A K 1 C14ce55_LED_A D3103 BAS16GW-QX
C14ce55_LED_K 2 A K 1 C14ce55_DRIVE D3104 BAS16GW-QX
C14ce55_LED_K 2 A K 1 C14ce55_SENSE D3105 BAS16GW-QX
C14ce55_LED_K 1 D109 2 C14ce55_LED_A APHHS1005QWF/D
IN_BUFFER 3 + U3103A 1 C14ce55_DRIVE
C14ce55_SENSE 2 − OPA4196IDR
AGND 5 + U3103B 7 LED_PACK_OPA4196IDR_0_IO_UNUSED2
LED_PACK_OPA4196IDR_0_IO_UNUSED2 6 − OPA4196IDR
AGND 10 + U3103C 8 LED_PACK_OPA4196IDR_0_IO_UNUSED3
LED_PACK_OPA4196IDR_0_IO_UNUSED3 9 − OPA4196IDR
AGND 12 + U3103D 14 LED_PACK_OPA4196IDR_0_IO_UNUSED4
LED_PACK_OPA4196IDR_0_IO_UNUSED4 13 − OPA4196IDR
+12V 4 V+ U3103E V− 11 -12V OPA4196IDR
+12V 1 C3108 2 AGND 100 nF
-12V 1 C3109 2 AGND 100 nF
IN_SENSE A 1 D3107 K 2 IN_SENSE BAT54S,215 COM IN_BUFFER

Sheet: /B1/
File: mult.kicad_sch

Title:

Size: A0 | Date: | **Rev:**

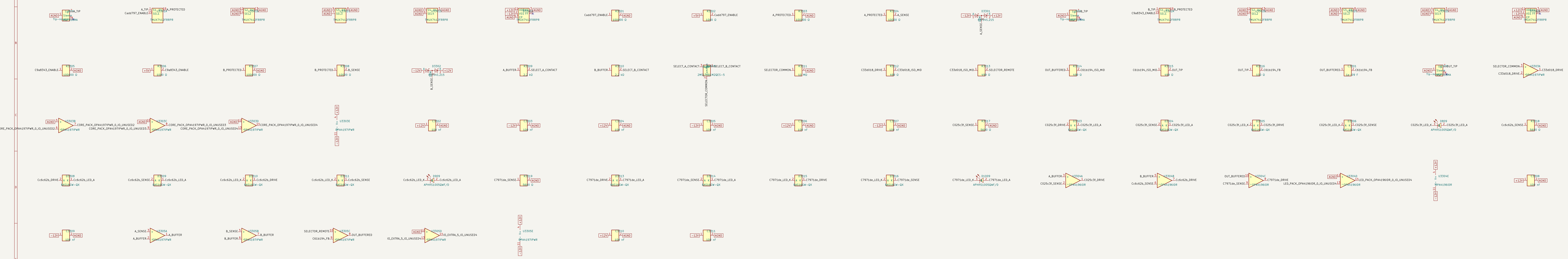
KiCad E.D.A. 10.0.6 | Id: 21/36



Sheet: /B2/
File: mult.kicad_sch

Title:

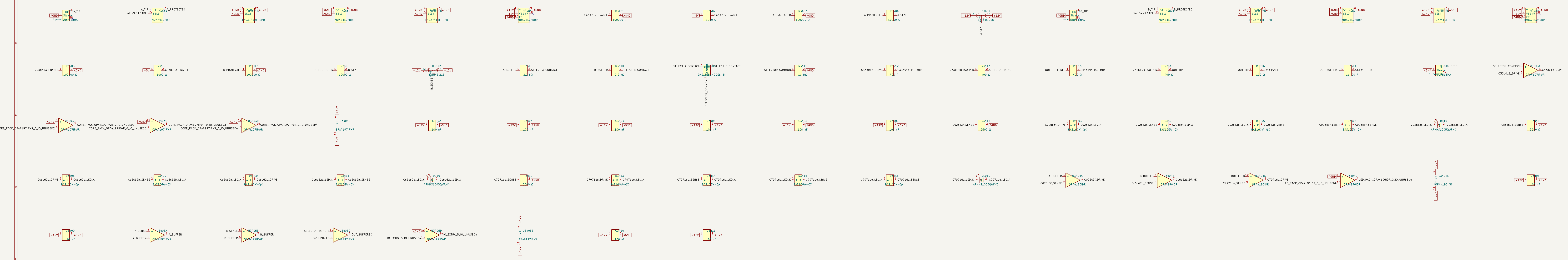
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 22/36



Sheet: /X1/
File: manual_ab.kicad_sch

Title:

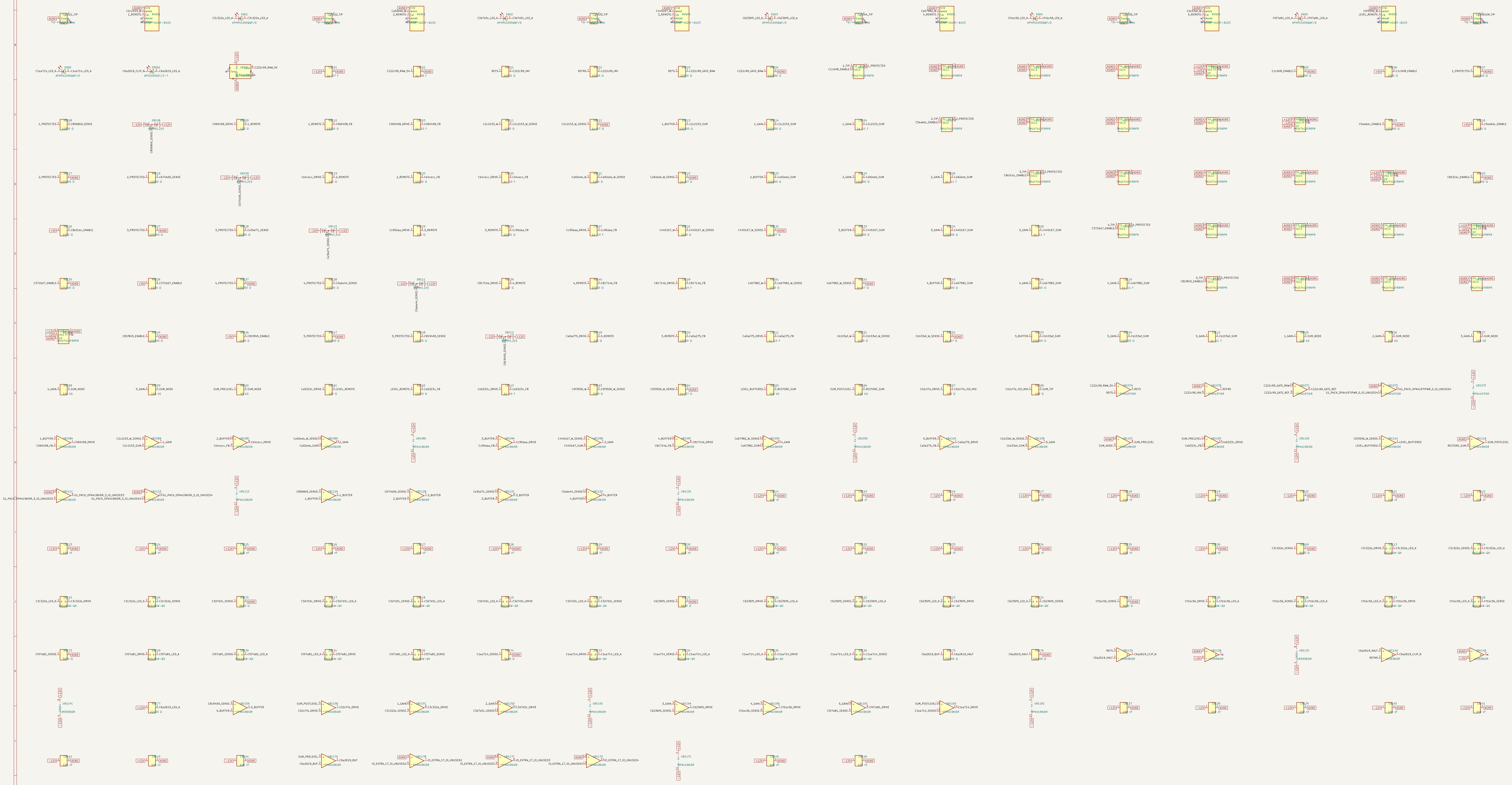
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 23/36



Sheet: /X2/
File: manual_ab.kicad_sch

Title:

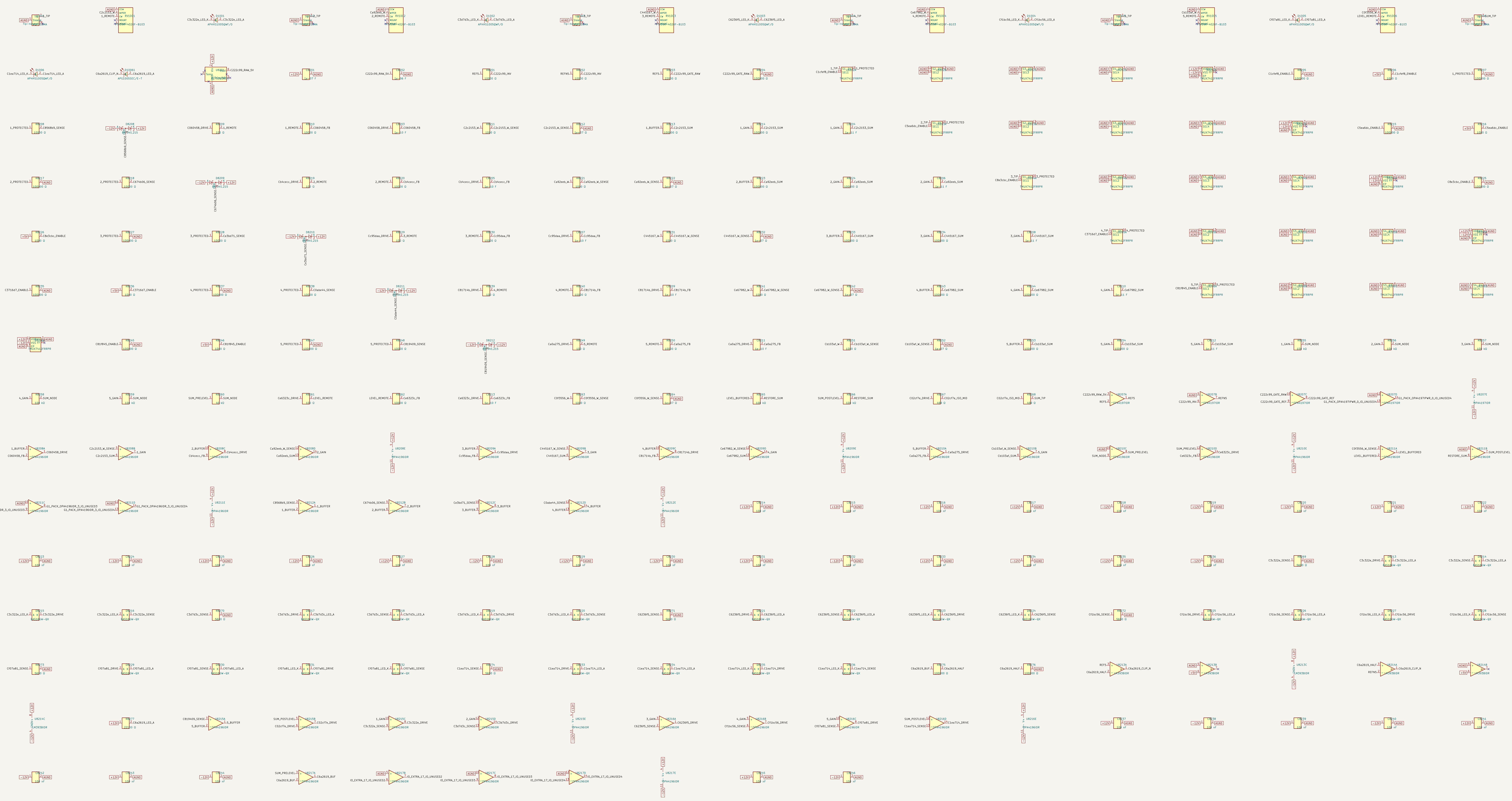
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 24/36



Sheet: /M5A/
File: mix5.kicad_sch

Title:

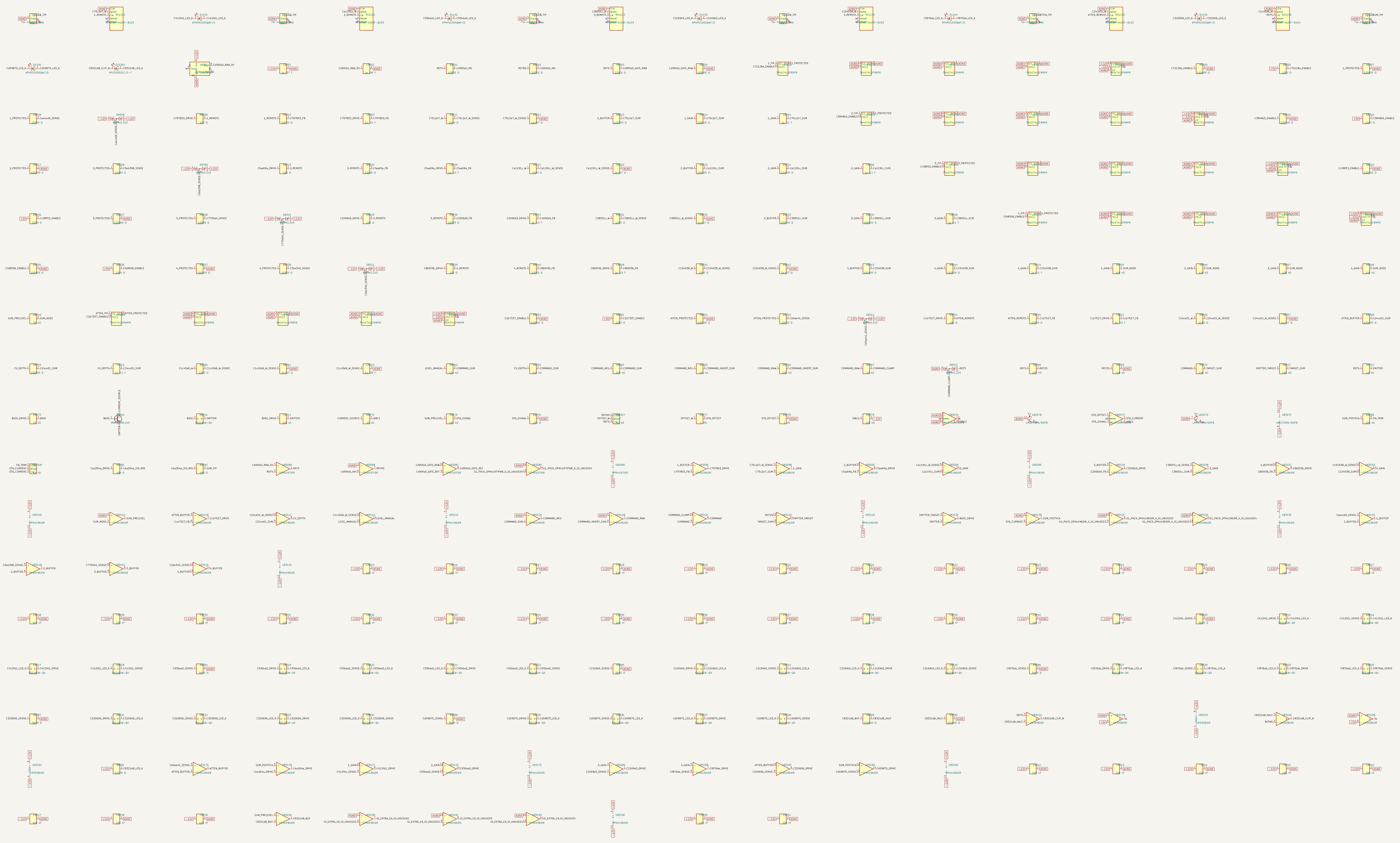
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 25/36



Sheet: /M5B/
File: mix5.kicad_sch

Title:

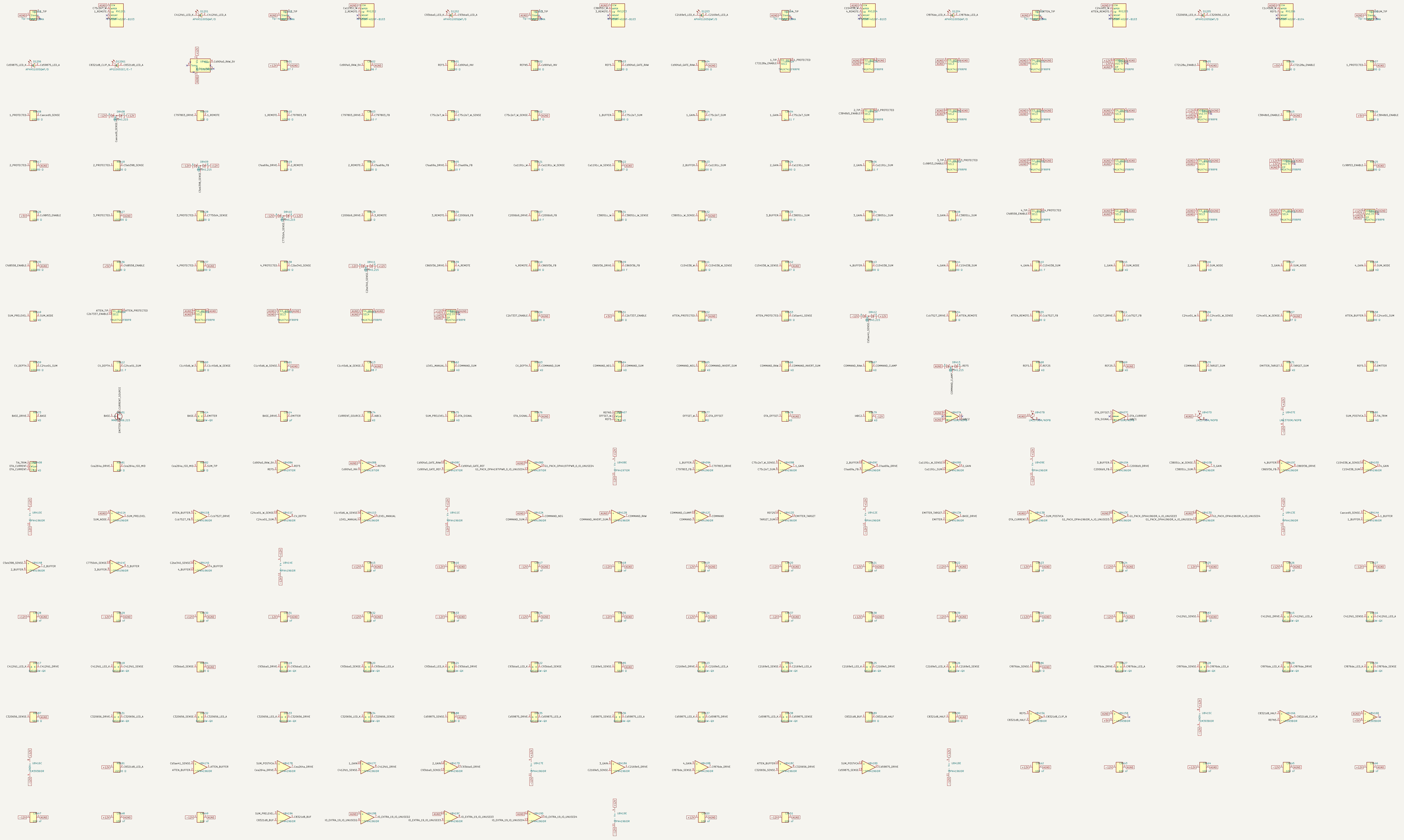
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 26/36



Sheet: /M4A/
File: mix4_vca.kicad_sch

Title:

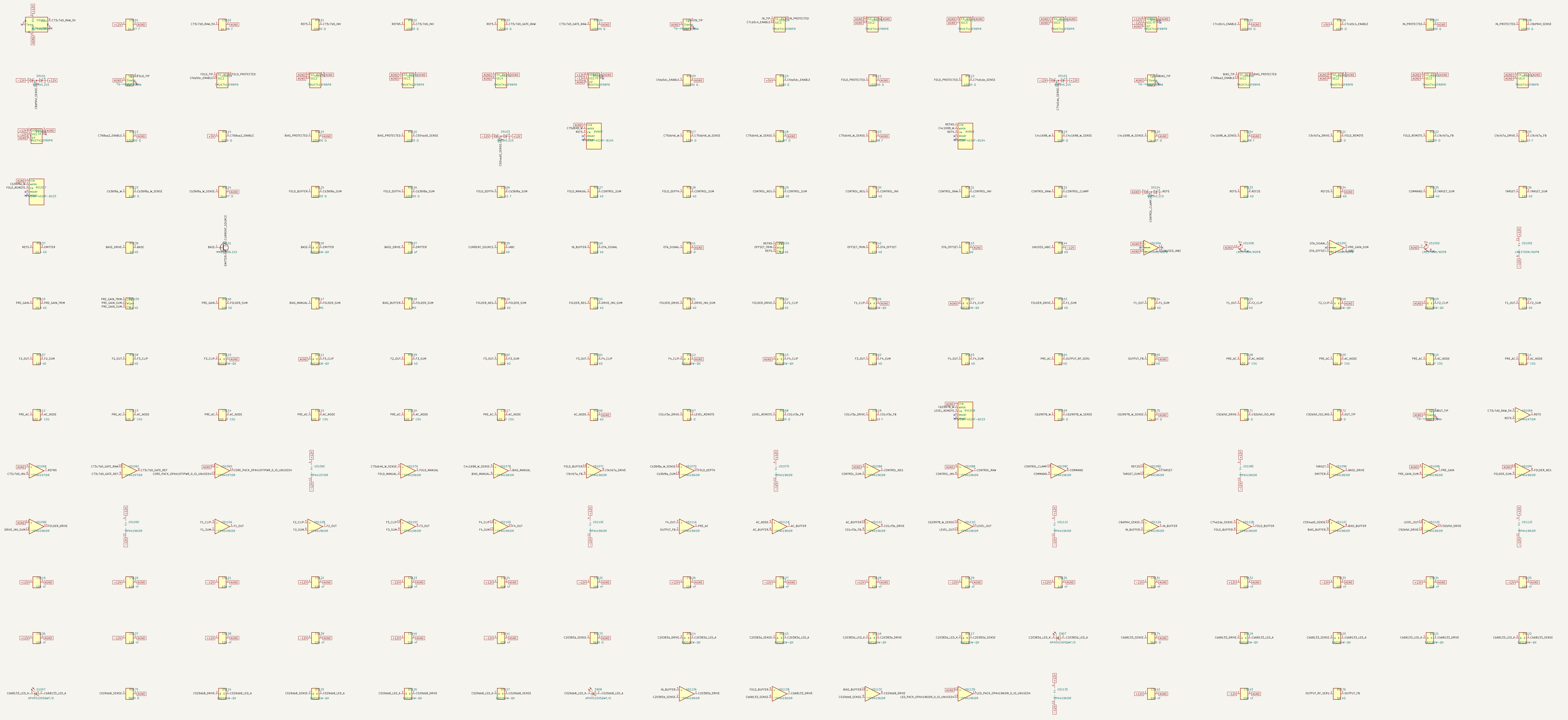
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 27/36



Sheet: /M4B/
File: mix4_vca.kicad_sch

Title:

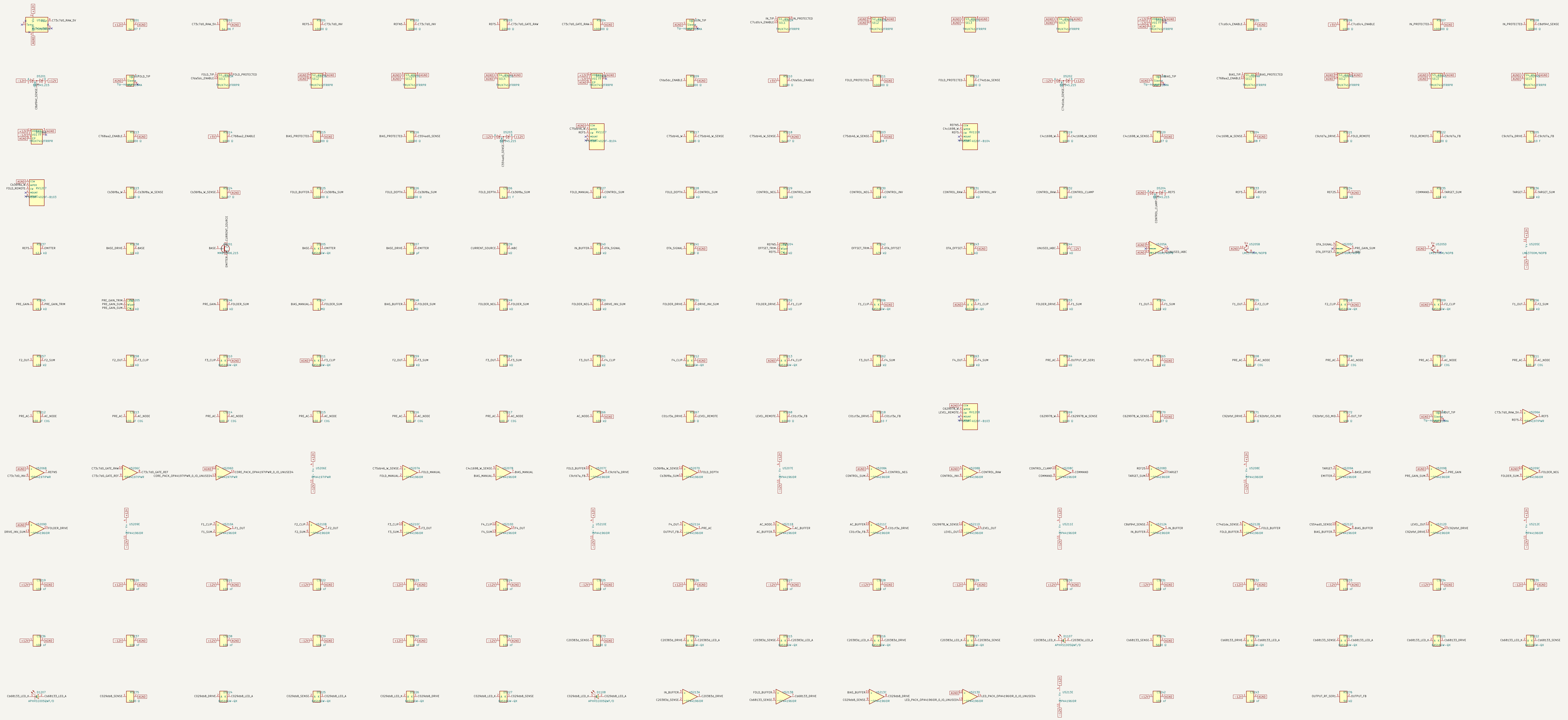
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 28/36



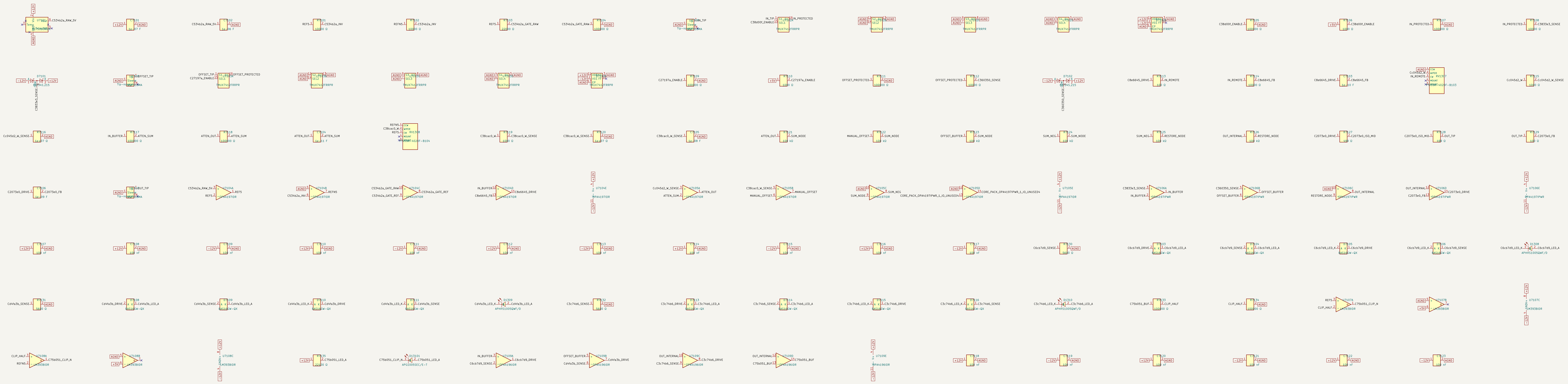
Sheet: /W2/
File: wavefolder.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 29/36



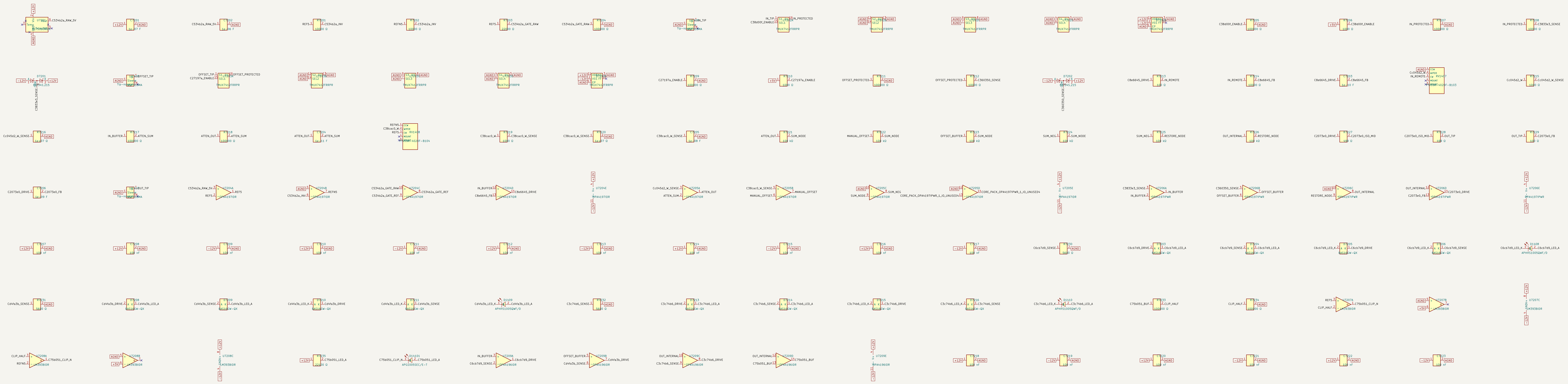
Sheet: /W1/
File: wavefolder_tssop.kicad_sch
Title:
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 30/36



Sheet: /A01/
File: offset.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 31/36



Sheet: /A02/
File: offset.kicad_sch

Title:

Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6

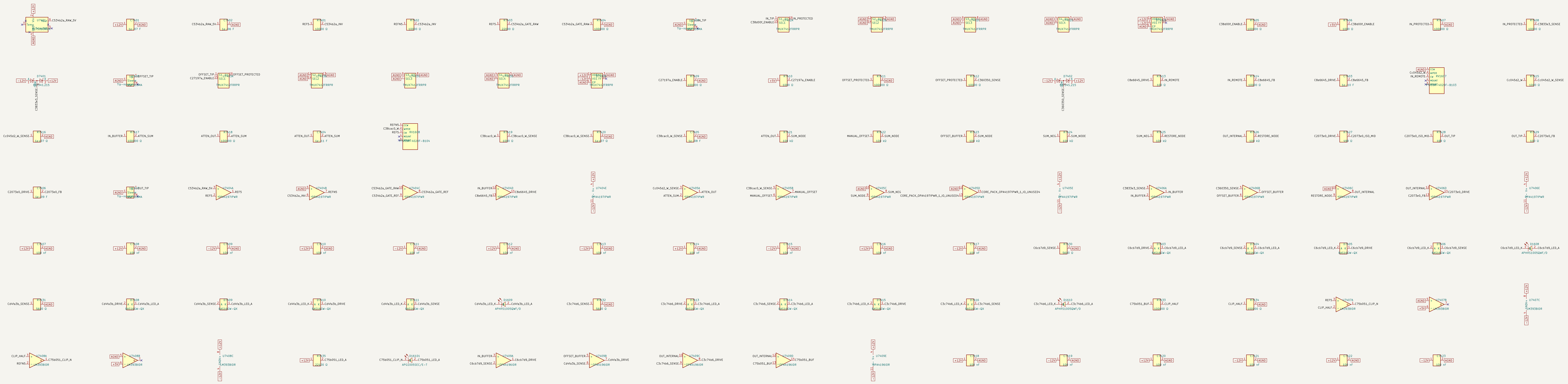
Id: 32/36

U7301 REF5050AIDR Vin Vout Temp GND Trim/NR +12V AGND C534b2a_RAW_5V
C7301 1e-07 F +12V AGND
C7302 1e-06 F C534b2a_RAW_5V AGND
R7301 10000 Ω REF5 C534b2a_INV
R7302 10000 Ω REFN5 C534b2a_INV
R7303 22000 Ω REF5 C534b2a_GATE_RAW
R7304 100000 Ω C534b2a_GATE_RAW AGND
J1508 WQP518MA Tip Sleeve TN Tip-normal AGND IN_TIP
U7302A TMUX7412FRRPR S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7302B TMUX7412FRRPR S2 D2 SEL2 AGND AGND AGND
U7302C TMUX7412FRRPR S3 D3 SEL3 AGND AGND AGND
U7302D TMUX7412FRRPR S4 D4 SEL4 AGND AGND AGND
U7302E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND AGND
R7305 100000 Ω C38d00f_ENABLE AGND
R7306 1000 Ω +5V C38d00f_ENABLE
R7307 100000 Ω IN_PROTECTED AGND
R7308 10000 Ω IN_PROTECTED C3833e3_SENSE
D7301 BAT54S,215 A K COM −12V +12V C3833e3_SENSE
J1509 WQP518MA Tip Sleeve TN Tip-normal AGND OFFSET_TIP
U7303A TMUX7412FRRPR S1 D1 SEL1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7303B TMUX7412FRRPR S2 D2 SEL2 AGND AGND AGND
U7303C TMUX7412FRRPR S3 D3 SEL3 AGND AGND AGND
U7303D TMUX7412FRRPR S4 D4 SEL4 AGND AGND AGND
U7303E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND AGND
R7309 100000 Ω C27197a_ENABLE AGND
R7310 1000 Ω +5V C27197a_ENABLE
R7311 100000 Ω OFFSET_PROTECTED AGND
R7312 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7302 BAT54S,215 A K COM −12V +12V C360350_SENSE
R7313 100 Ω C8e6645_DRIVE IN_REMOTE
R7314 10000 Ω IN_REMOTE C8e6645_FB
C7303 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1507 PTV09A-4020F-B103 CCW WIPER CW MOUNT MOUNT AGND Cc045d2_W IN_REMOTE
R7315 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7316 1e+07 Ω Cc045d2_W_SENSE AGND
R7317 100000 Ω IN_BUFFER ATTEN_SUM
R7318 100000 Ω ATTEN_OUT ATTEN_SUM
C7304 1e-11 F ATTEN_OUT ATTEN_SUM
RV1508 PTV09A-4020F-B104 CCW WIPER CW MOUNT MOUNT REFN5 C38cac0_W REF5
R7319 1000 Ω C38cac0_W C38cac0_W_SENSE
R7320 1e+07 Ω C38cac0_W_SENSE AGND
C7305 1e-08 F C38cac0_W_SENSE AGND
R7321 100 kΩ ATTEN_OUT SUM_NODE
R7322 100 kΩ MANUAL_OFFSET SUM_NODE
R7323 100 kΩ OFFSET_BUFFER SUM_NODE
R7324 100 kΩ SUM_NEG SUM_NODE
R7325 100 kΩ SUM_NEG RESTORE_NODE
R7326 100 kΩ OUT_INTERNAL RESTORE_NODE
R7327 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7328 499 Ω C2073e0_ISO_MID OUT_TIP
R7329 100 Ω OUT_TIP C2073e0_FB
C7306 1e-09 F C2073e0_DRIVE C2073e0_FB
J1510 WQP518MA Tip Sleeve TN Tip-normal AGND OUT_TIP
U7304A OPA4197IDR C534b2a_RAW_5V REF5 REF5
U7304B OPA4197IDR AGND C534b2a_INV REFN5
U7304C OPA4197IDR C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF
U7304D OPA4197IDR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7304E OPA4197IDR V+ V− +12V −12V
U7305A OPA4197IDR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7305B OPA4197IDR C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET
U7305C OPA4197IDR AGND SUM_NODE SUM_NEG
U7305D OPA4197IDR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7305E OPA4197IDR V+ V− +12V −12V
U7306A OPA4197IPWR C3833e3_SENSE IN_BUFFER IN_BUFFER
U7306B OPA4197IPWR C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER
U7306C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7306D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7306E OPA4197IPWR V+ V− +12V −12V
C7307 100 nF +12V AGND
C7308 100 nF +12V AGND
C7309 100 nF −12V AGND
C7310 100 nF +12V AGND
C7311 100 nF −12V AGND
C7312 100 nF +12V AGND
C7313 100 nF −12V AGND
C7314 100 nF +12V AGND
C7315 100 nF −12V AGND
C7316 100 nF +12V AGND
C7317 100 nF −12V AGND
R7330 5600 Ω C6cb7d9_SENSE AGND
D7303 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7304 BAS16GW-QX A K C6cb7d9_SENSE C6cb7d9_LED_A
D7305 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_DRIVE
D7306 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_SENSE
D1508 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7331 5600 Ω Cd4fa3b_SENSE AGND
D7308 BAS16GW-QX A K Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7309 BAS16GW-QX A K Cd4fa3b_SENSE Cd4fa3b_LED_A
D7310 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7311 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_SENSE
D1509 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7332 5600 Ω C3c74b6_SENSE AGND
D7313 BAS16GW-QX A K C3c74b6_DRIVE C3c74b6_LED_A
D7314 BAS16GW-QX A K C3c74b6_SENSE C3c74b6_LED_A
D7315 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_DRIVE
D7316 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_SENSE
D1510 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7333 100000 Ω C75b051_BUF CLIP_HALF
R7334 100000 Ω CLIP_HALF AGND
U7307A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7307B LM393BIDR AGND +5V
U7307C LM393BIDR GND V+ +12V −12V
U7308A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7308B LM393BIDR AGND +5V
U7308C LM393BIDR GND V+ +12V −12V
R7335 22100 Ω +12V C75b051_LED_A
D15101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7309A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7309B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7309C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7309D OPA4196IDR OUT_INTERNAL C75b051_BUF C75b051_BUF
U7309E OPA4196IDR V+ V− +12V −12V
C7318 100 nF +12V AGND
C7319 100 nF −12V AGND
C7320 100 nF +12V AGND
C7321 100 nF −12V AGND
C7322 100 nF +12V AGND
C7323 100 nF −12V AGND

Sheet: /A03/
File: offset.kicad_sch

Title:

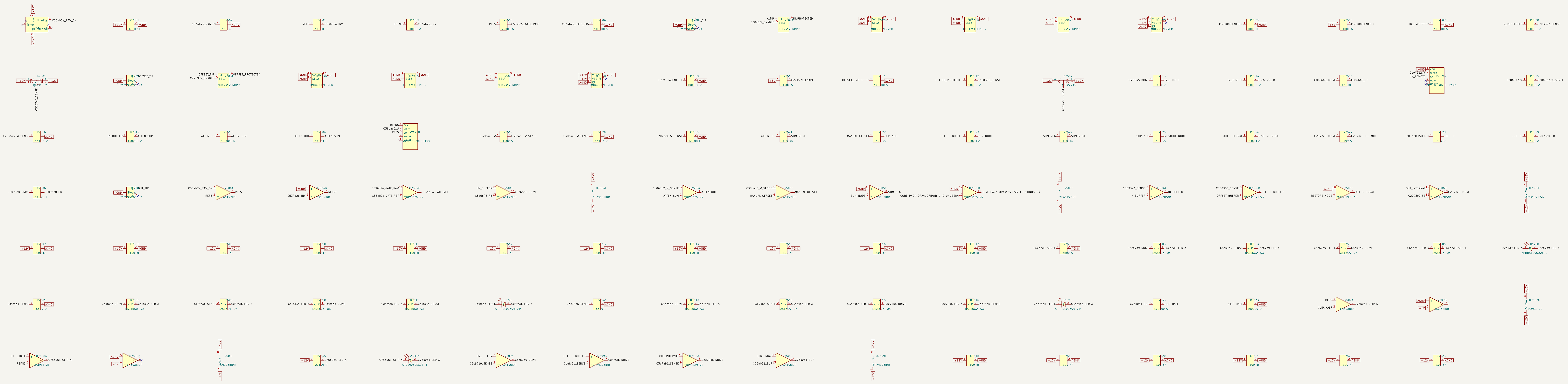
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 33/36



Sheet: /A04/
File: offset_tssop.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 34/36

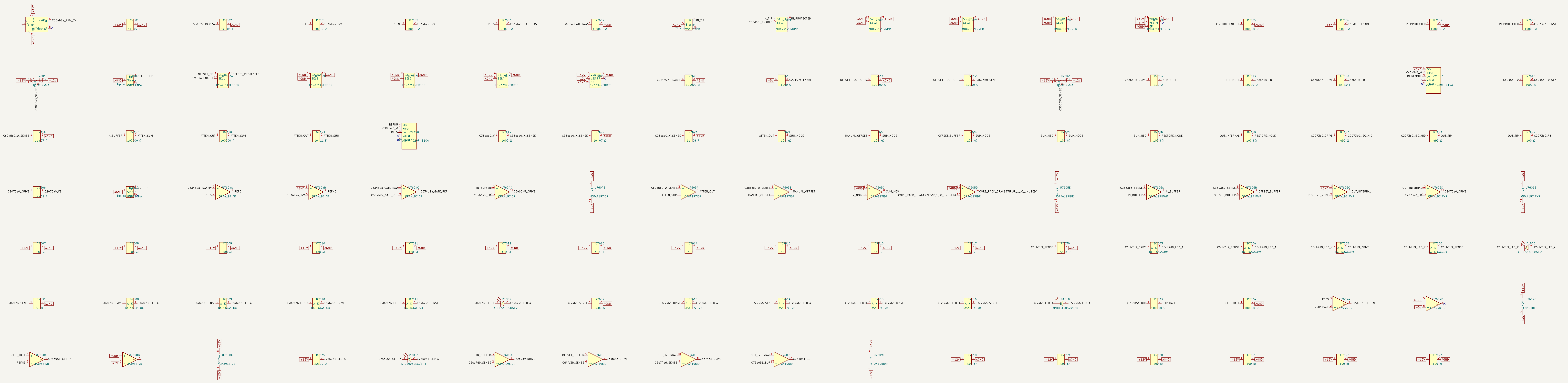


Sheet: /A05/
File: offset.kicad_sch

Title:

Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 35/36


U7601 REF5050AIDR +12V AGND C534b2a_RAW_5V Vin Vout GND Temp Trim/NR
C7601 +12V AGND 1e-07 F
C7602 C534b2a_RAW_5V AGND 1e-06 F
R7601 REF5 C534b2a_INV 10000 Ω
R7602 REFN5 C534b2a_INV 10000 Ω
R7603 REF5 C534b2a_GATE_RAW 22000 Ω
R7604 C534b2a_GATE_RAW AGND 100000 Ω
IN_TIP AGND Tip Sleeve TN Tip-normal WQP-518MA
U7602A IN_TIP C38d00f_ENABLE IN_PROTECTED S1 D1 SEL1 TMUX7412FRRPR
U7602B AGND S2 D2 SEL2 TMUX7412FRRPR
U7602C AGND S3 D3 SEL3 TMUX7412FRRPR
U7602D AGND S4 D4 SEL4 TMUX7412FRRPR
U7602E +12V −12V AGND VDD GND VSS EP FF TMUX7412FRRPR
R7605 C38d00f_ENABLE AGND 100000 Ω
R7606 +5V C38d00f_ENABLE 1000 Ω
R7607 IN_PROTECTED AGND 100000 Ω
R7608 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7601 −12V +12V A K BAT54S,215 C3833e3_SENSE
AGND OFFSET_TIP Tip Sleeve TN Tip-normal WQP-518MA
U7603A OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED S1 D1 SEL1 TMUX7412FRRPR
U7603B AGND S2 D2 SEL2 TMUX7412FRRPR
U7603C AGND S3 D3 SEL3 TMUX7412FRRPR
U7603D AGND S4 D4 SEL4 TMUX7412FRRPR
U7603E +12V −12V AGND VDD GND VSS EP FF TMUX7412FRRPR
R7609 C27197a_ENABLE AGND 100000 Ω
R7610 +5V C27197a_ENABLE 1000 Ω
R7611 OFFSET_PROTECTED AGND 100000 Ω
R7612 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7602 −12V +12V A K BAT54S,215 C360350_SENSE
R7613 C8e6645_DRIVE IN_REMOTE 100 Ω
R7614 IN_REMOTE C8e6645_FB 10000 Ω
C7603 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1807 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT PTV09A-4020F-B103
R7615 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7616 Cc045d2_W_SENSE AGND 1e+07 Ω
R7617 IN_BUFFER ATTEN_SUM 100000 Ω
R7618 ATTEN_OUT ATTEN_SUM 100000 Ω
C7604 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1808 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT PTV09A-4020F-B104
R7619 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7620 C38cac0_W_SENSE AGND 1e+07 Ω
C7605 C38cac0_W_SENSE AGND 1e-08 F
R7621 ATTEN_OUT SUM_NODE 100 kΩ
R7622 MANUAL_OFFSET SUM_NODE 100 kΩ
R7623 OFFSET_BUFFER SUM_NODE 100 kΩ
R7624 SUM_NEG SUM_NODE 100 kΩ
R7625 SUM_NEG RESTORE_NODE 100 kΩ
R7626 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7627 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7628 C2073e0_ISO_MID OUT_TIP 499 Ω
R7629 OUT_TIP C2073e0_FB 100 Ω
C7606 C2073e0_DRIVE C2073e0_FB 1e-09 F
AGND OUT_TIP Tip Sleeve TN Tip-normal WQP-518MA
U7604A C534b2a_RAW_5V REF5 REF5 OPA4197IDR
U7604B AGND C534b2a_INV REFN5 OPA4197IDR
U7604C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IDR
U7604D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IDR
U7604E +12V −12V V+ V− OPA4197IDR
U7605A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IDR
U7605B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IDR
U7605C AGND SUM_NODE SUM_NEG OPA4197IDR
U7605D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IDR
U7605E +12V −12V V+ V− OPA4197IDR
U7606A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7606B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7606C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7606D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7606E +12V −12V V+ V− OPA4197IPWR
C7607 +12V AGND 100 nF
C7608 +12V AGND 100 nF
C7609 −12V AGND 100 nF
C7610 +12V AGND 100 nF
C7611 −12V AGND 100 nF
C7612 +12V AGND 100 nF
C7613 −12V AGND 100 nF
C7614 +12V AGND 100 nF
C7615 −12V AGND 100 nF
C7616 +12V AGND 100 nF
C7617 −12V AGND 100 nF
R7630 C6cb7d9_SENSE AGND 5600 Ω
D7603 C6cb7d9_DRIVE C6cb7d9_LED_A A K BAS16GW-QX
D7604 C6cb7d9_SENSE C6cb7d9_LED_A A K BAS16GW-QX
D7605 C6cb7d9_LED_K C6cb7d9_DRIVE A K BAS16GW-QX
D7606 C6cb7d9_LED_K C6cb7d9_SENSE A K BAS16GW-QX
D1808 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7631 Cd4fa3b_SENSE AGND 5600 Ω
D7608 Cd4fa3b_DRIVE Cd4fa3b_LED_A A K BAS16GW-QX
D7609 Cd4fa3b_SENSE Cd4fa3b_LED_A A K BAS16GW-QX
D7610 Cd4fa3b_LED_K Cd4fa3b_DRIVE A K BAS16GW-QX
D7611 Cd4fa3b_LED_K Cd4fa3b_SENSE A K BAS16GW-QX
D1809 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7632 C3c74b6_SENSE AGND 5600 Ω
D7613 C3c74b6_DRIVE C3c74b6_LED_A A K BAS16GW-QX
D7614 C3c74b6_SENSE C3c74b6_LED_A A K BAS16GW-QX
D7615 C3c74b6_LED_K C3c74b6_DRIVE A K BAS16GW-QX
D7616 C3c74b6_LED_K C3c74b6_SENSE A K BAS16GW-QX
D1810 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7633 C75b051_BUF CLIP_HALF 100000 Ω
R7634 CLIP_HALF AGND 100000 Ω
U7607A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7607B AGND +5V LM393BIDR
U7607C +12V −12V V+ GND LM393BIDR
U7608A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7608B AGND +5V LM393BIDR
U7608C +12V −12V V+ GND LM393BIDR
R7635 +12V C75b051_LED_A 22100 Ω
D18101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7609A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7609B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7609C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7609D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7609E +12V −12V V+ V− OPA4196IDR
C7618 +12V AGND 100 nF
C7619 −12V AGND 100 nF
C7620 +12V AGND 100 nF
C7621 −12V AGND 100 nF
C7622 +12V AGND 100 nF
C7623 −12V AGND 100 nF

Sheet: /A06/
File: offset.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 36/36